MAT 2040 Linear Algebra

Assignment 4

- Released date: 2024/06/24.
- Due: 2024/06/27.
- Late submission is NOT acceptable.
- Please submit your assignment as a PDF file titled "student ID + HW4".

Question 1. Determine whether the following sets are bases for $\mathbb{R}^3$.

(a)
$$\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}$$

(b)
$$\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}$$

(c)
$$\begin{bmatrix}1\\0\\-2\end{bmatrix}, \begin{bmatrix}3\\2\\-4\end{bmatrix}, \begin{bmatrix}-3\\-5\\1\end{bmatrix}$$

Question 2. Find a basis for the solution set of the following system:
$$\begin{bmatrix}-2 & 4 & -2 & -4\\ 2 & -6 & -3 & 1\\ -3 & 8 & 2 & -3\end{bmatrix}\mathbf{x} = \mathbf{0}$$

Question 3. In each of the following, find the dimension of the subspace of P_3 spanned by the given vectors:

(a) $x, x-1, x^2+1$

(b) $x, x-1, x^2+1, x^2-1$

Question 4. Let $\{x, 1\}$ and $\{2x-1, 2x+1\}$ be ordered bases for P_2, find the transition matrix representing the change in coordinates from $\{2x-1, 2x+1\}$ to $\{x, 1\}$.

Question 5. Given $\mathbf{x}_1=(1,1,1)^T$ and $\mathbf{x}_2=(3,-1,4)^T$:

(a) Do $\mathbf{x}_1$ and $\mathbf{x}_2$ span $\mathbb{R}^3$? Explain.

(b) Let $\mathbf{x}_3$ be a third vector in $\mathbb{R}_3$ and set $X = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$. What condition(s) would X have to satisfy in order for $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ to form a basis for $\mathbb{R}_3$?

(c) Find a third vector $\mathbf{x}_3$ that will extend the set $\{\mathbf{x}_1, \mathbf{x}_2\}$ to a basis for $\mathbb{R}^3$.

Question 6. The vectors $\mathbf{x}_1 = \begin{bmatrix}1\\2\\2\end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix}2\\5\\4\end{bmatrix}, \mathbf{x}_3 = \begin{bmatrix}1\\3\\2\end{bmatrix}, \mathbf{x}_4 = \begin{bmatrix}2\\7\\4\end{bmatrix}, \mathbf{x}_5 = \begin{bmatrix}1\\1\\0\end{bmatrix}$ span $\mathbb{R}^3$. Pare down the set $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5\}$ to form a basis for $\mathbb{R}^3$.

Question 7. Let

$$u_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}, u_2 = \begin{bmatrix}1\\2\\2\end{bmatrix}, u_3 = \begin{bmatrix}2\\3\\4\end{bmatrix}, v_1 = \begin{bmatrix}4\\6\\7\end{bmatrix}, v_2 = \begin{bmatrix}0\\1\\1\end{bmatrix}, v_3 = \begin{bmatrix}0\\1\\2\end{bmatrix},$$

(a) Find the transition matrix from $\{v_1, v_2, v_3\}$ to $\{u_1, u_2, u_3\}$

(b) If $\mathbf{x} = 2v_1 + 3v_2 - 4v_3$, determine the coordinate of $\mathbf{x}$ with respect to $\{u_1, u_2, u_3\}$.

Question 8. Let $B = \left\{\begin{bmatrix}7\\5\end{bmatrix}, \begin{bmatrix}-3\\-1\end{bmatrix}\right\}$, $C = \left\{\begin{bmatrix}1\\-5\end{bmatrix}, \begin{bmatrix}-2\\2\end{bmatrix}\right\}$ be bases for $\mathbb{R}^2$. Find the transition matrix from B to C and the transition matrix from C to B.

Question 9. Let $b_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, b_2 = \begin{bmatrix}-3\\4\\0\end{bmatrix}, b_3 = \begin{bmatrix}3\\-6\\3\end{bmatrix}$.

(a) Show that the set $\mathbf{B} = \{b_1, b_2, b_3\}$ is a basis of $\mathbb{R}^3$.

(b) Given a vector $\mathbf{x} = \begin{bmatrix}-8\\2\\3\end{bmatrix}$ in $\mathbb{R}^3$. Find the coordinate vector of $\mathbf{x}$ with respect to the basis $\mathbf{B}$.

(c) Given the coordinate of $\mathbf{y}$ with respect to the basis $\mathbf{B}$ is $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. What is the vector $\mathbf{y}$ in $\mathbb{R}^3$.

Question 10. Let U and V be subspaces of a vector space W.
Define $U + V = \{\mathbf{z}|\mathbf{z}=\mathbf{u}+\mathbf{v}\ \textit{where}\ \mathbf{u} \in U\ \textit{and}\ \mathbf{v} \in V\}$. Show that $U + V$ is a subspace of W.

Question 11. In $\mathbb{R}^4$, let U be the subspace of all vectors of the form $(u_1, u_2, 0, 0)^T$, and let V be the subspace of all vectors of the form $(0, v_2, v_3, 0)^T$. What are the dimensions of $U, V, U \cap V, U + V$? Find a basis for each of these four subspaces.

Question 12. Show that if U and V are subspaces of $\mathbb{R}^n$ and $U \cap V = \{\mathbf{0}\}$, then dim $(U + V)$ = dim U+ dim V.